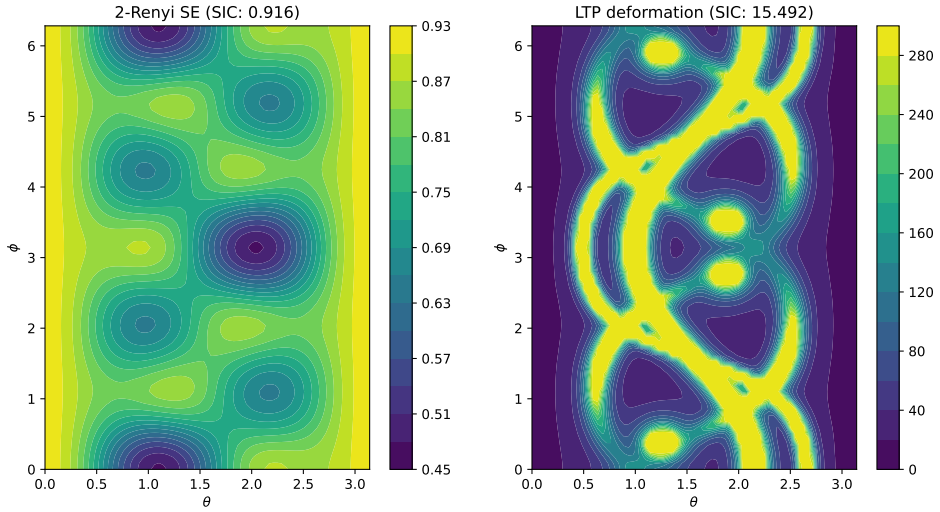
A Qubit from Two Compound SIC Fiducials

The notebook `CompoundSICQubitViz.ipynb` considers the qubit spanned by the two vectors `sachin[0][0]` and `sachin[1][0]`, and visualizes the stabilizer entropy and LTP deformation across the Bloch sphere.



Here we give an explicit expression for its pure states. Letting

$$a = \sqrt{2+\sqrt{5}}, \qquad b = \sqrt{\frac{1-1/\sqrt{5}}{8}}, \qquad q = e^{i\pi/4} = \frac{1+i}{\sqrt{2}}, \tag{1}$$

we have for the two chosen SIC fiducials,

$$|f_0\rangle = b\begin{pmatrix} 1+q \\ i(1+aq) \\ q-1 \\ i(1-aq) \end{pmatrix}, \qquad |f_1\rangle = b\begin{pmatrix} -a-q \\ i(1+q) \\ a-q \\ i(1-q) \end{pmatrix}. \tag{2}$$

These are orthonormal. Indeed, using $|q| = 1$ and $q + q^* = \sqrt{2}$,

$$\langle f_0|f_0\rangle = \langle f_1|f_1\rangle = b^2(6+2a^2) = 1, \tag{3}$$

$$\langle f_0|f_1\rangle = 0. \tag{4}$$

Thus the general pure state of this encoded qubit is

$$|\psi(\theta,\phi)\rangle = \cos\frac{\theta}{2}\,|f_0\rangle + e^{i\phi}\sin\frac{\theta}{2}\,|f_1\rangle, \qquad 0 \le \theta \le \pi, \quad 0 \le \phi < 2\pi. \tag{5}$$

Writing $c = \cos(\theta/2)$ and $s = \sin(\theta/2)$, this becomes

$$\boxed{|\psi(\theta,\phi)\rangle = b \begin{pmatrix} (1+q)c - (a+q)e^{i\phi}s \\ i\big[(1+aq)c + (1+q)e^{i\phi}s\big] \\ (q-1)c + (a-q)e^{i\phi}s \\ i\big[(1-aq)c + (1-q)e^{i\phi}s\big] \end{pmatrix}}. \tag{6}$$